

The Evolution of Agentic AI: From Prediction to Action

| Moving from Generative Tools to Autonomous AI Workers



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Faculty Presentation Series

| What is Artificial Intelligence?

Natural vs. Artificial

Natural intelligence is the unique human capacity to observe scenarios, apply context, synthesize experiences, and make custom decisions.

Artificial Intelligence mimics this capability by programming structured rules, heuristics, and deep models into computational machines.

The Microwave Paradigm Shift

***Traditional Approach:** A human reviews a frozen meal, estimates a safe defrost time, and manually inputs a "5-minute" command timer.*

AI-Powered Approach: The microwave scans the thermal profile of the food directly, detects density, and adjusts heating steps autonomously.

| The Three Levels of AI



LEVEL 1

Predictive AI

Dominant pre-2022. Designed for analytical pattern matching and labeling. Produces static outputs like "cat/dog classification" or "spam detection" without creating anything new.



LEVEL 2

Generative AI

2022 to present. Large models (e.g., ChatGPT) trained to synthesize new text, code, or images based directly on a user-submitted conversational prompt.



LEVEL 3

Agentic AI

The emerging frontier. Action and goal-oriented software. It autonomously plans, tools-up, researches, tests, and completes complex sequences from beginning to end.

| Key Characteristics of Agentic AI



Structured Planning

The system digests an ambiguous, high-level goal and maps it into a series of logical, step-by-step instructions to prevent execution errors.



Adaptive Tool Usage

Leverages outer-world integrations (APIs, web browsers, local files, execution terminals) to gather context and compute values dynamically.



Autonomous Self-Correction

Continuously critiques its own output, reviews errors in runtime, and iterates automatically without throwing exceptions back to the human.



Outcome & Action Focus

While Generative AI satisfies a query with chat text, Agentic AI executes procedures to deliver a completed final outcome.

I AI as a Tool vs. AI as a Worker

AI AS A TOOL

The Mechanical Drill Analogy

The AI is treated as a highly functional, specialized instrument. It is incapable of working without constant, active human operation. The human remains the sole project driver, doing the strategic thinking and manually commanding every sub-task.

AI AS A WORKER

The External Contractor Analogy

The AI functions as an autonomous hire. The human sets the high-level intent, constraints, and standard of quality. The AI Worker assumes management of the timeline and executes the micro-details under the human's strategic direction.

| The "10-80-10 Rule"

PHASE 01 — FIRST 10%

Human Direction

The human sets the specific scope, defines boundaries, specifies targets, and inputs initial high-intent parameters.

PHASE 02 — MIDDLE 80%

AI Execution

Autonomous AI Workers carry out the heavy research, testing, assembly, writing, and logical debugging cycles.

PHASE 03 — FINAL 10%

Human Validation

The human reviews, acts as an editor, modifies details for branding, and takes legal and operational responsibility.

| Human Role: Promoted, Not Replaced

AI absorbs the mediocre middle tasks, freeing humans to scale upward.

Instead of replacing workforce structures, Agentic AI upgrades professional roles, pushing humans into positions of strategic supervision.

Humans keep the most critical assets: Goal Direction and Judgment.

~~Typist~~



Editor

~~Coder~~



Architect

~~Worker~~



Director

| The Autonomous Agent Factory

190

AI WORKERS / 10 HUMANS

The "Agent Factory" Paradigm

A standardized industrial approach to Design, Develop, Ship, and Monitor dedicated AI worker nodes that operate securely in production environments.

AI-Native Organizations

Businesses where fundamental verticals (Sales, Marketing, Dev, Ops) are manned by specialized AI Agents under the steering oversight of small, elite human teams.

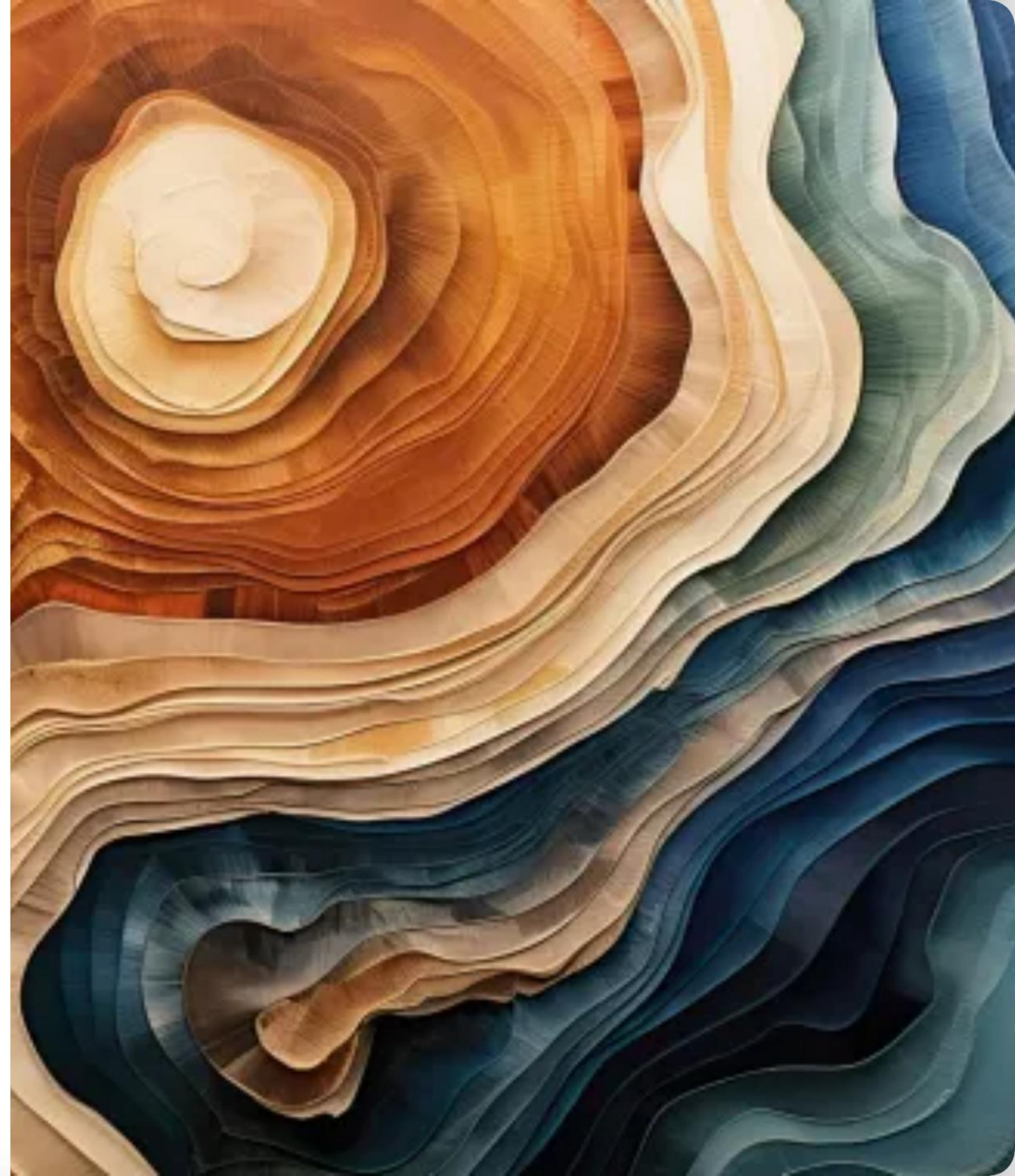
| The Personal AI Delegate

Solving the human bottleneck with a Digital Twin

Humans cannot coordinate, track, and monitor dozens of highly specialized sub-agents running at lightspeed.

The solution is your **Digital Twin (Personal Delegate)**. This primary node mirrors your professional style, tone, logic, and operational boundaries.

It directs downstream sub-agents and summarizes results for your quick approval.



| Seven Golden Rules for AI

1

Human-in-the-Loop

Humans drive initial intent (10%) and provide final strategic verification (10%).

2

The Personal Delegate

Every human coordinates specialized worker nodes through a personal digital twin.

3

Management Hierarchy

AI managers coordinate sub-agents dynamically based on real-time demands.

4

Task-Engine Fit

Match the computing power and model size precisely to the complexity of the task.

5

Unified Systems of Record

Agents must read from and write back to live, unified organizational data stores.

6

Rule-Bounded Budgets

Define absolute financial limits, operational parameters, and strict sandboxes.

7

Integrated Agent Nervous System

Ensure low-latency, event-driven, inter-agent communication channels exist completely independent of manual human intervention.

| Image Sources



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